$\vec{p} = m\vec{v}$ m – инерциальная масса $m \neq m$ (грав.)

$\vec{F} = \frac{d\vec{p}}{dt}$ Сила – мера взаимодействия двух тел, проявляющаяся в изменении импульса.

$\Downarrow$

$\frac{d\vec{v}}{dt} = \frac{\vec{F}}{m}$ Масса – мера инертности тела

При столкновении двух тел: $\left|\frac{\Delta v_1}{\Delta v_2}\right| = \frac{m_2}{m_1} \Rightarrow m_1 \Delta \vec{v_1} = -m_2 \Delta v_2$

$\Delta(m_1 \vec{v_1}) = -\Delta(m_2 \vec{v_2})$

$\Delta \vec{p_1} = -\Delta \vec{p_2}$

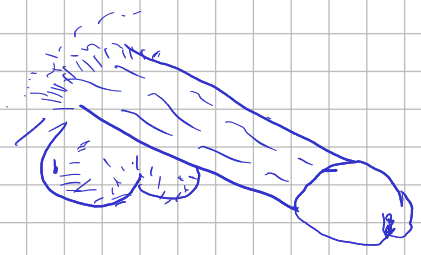
$\Delta \vec{p_1} + \Delta \vec{p_2} = 0$ – ЗСИ

только в замкнутых системах

В релятивистской механике:

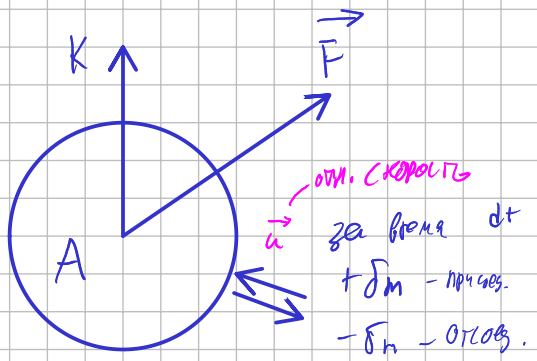
$\vec{p} = \frac{m\vec{v}}{\sqrt{1 - \frac{v^2}{c^2}}}$

$\vec{F_{12}} = -\vec{F_{21}}$



СИ	СГС
1кг	1г
1м	1см
1с	1с

При столкновении двух одинаковых шаров, один теряет импульс, а второй его приобретает

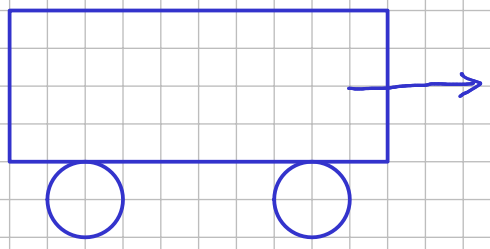


$m\,d\vec{v} = \vec{F}\,dt \pm \delta m\,\vec{u}$ ($\delta m \to dm$)

$m(t)\frac{d\vec{v}}{dt} = \vec{F} + \frac{dm}{dt}\vec{u}$ – Уравнение Мещерского

$\frac{dm}{dt}\vec{u}$ – реактивная сила

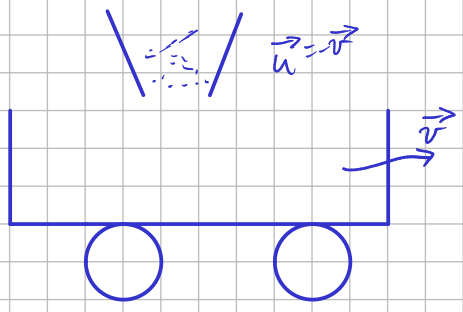
1



$\vec{u} = 0$

$m(t)\frac{d\vec{v}}{dt} = \vec{F}$

2



$m(t)\frac{d\vec{v}}{dt} = \vec{F} - \frac{dm}{dt}\vec{v}$ $\frac{d(mv)}{dt}$

$\frac{d(m\vec{v})}{dt} = \vec{F}$

③ Свободное пространство

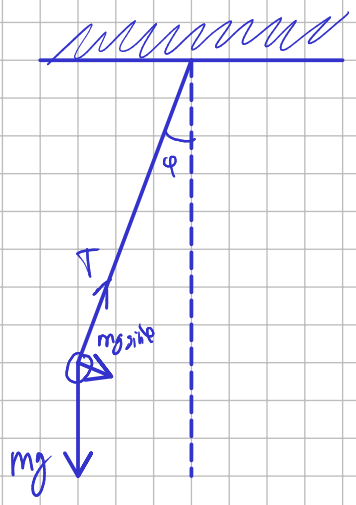
$$m\frac{dv}{dt} = -\underset{\text{const}}{u}\frac{dm}{dt}$$ против $\vec{v}$

$$u\int_{m_0}^{m(t)}\frac{dm}{m} = -\int_0^{v(t)} dv$$

$$u\ln\frac{m(t)}{m_0} = -v(t)$$

$$v(t) = u\ln\frac{m_0}{m(t)}$$

$$m(t) = m_0 e^{-\frac{v}{u}}$$



$\sin\varphi \approx \varphi$

$$F_{\text{бок}} = -mg\sin\varphi \approx -mg\varphi$$

$$ma_\tau = m\dot{\omega}l = m\ddot{\varphi}l = -mg\varphi$$

$$m\ddot{\varphi}l + mg\varphi = 0$$

$$\ddot{\varphi} + \underbrace{\frac{g}{l}}_{\omega^2\text{ - угл. частота}}\varphi = 0 \qquad T = 2\pi\sqrt{\frac{l}{g}}$$

$$\frac{dm}{dt} = \dot{m} = 2\underset{[\frac{1}{c}]}{\delta}m$$

$$m\frac{dv}{dt} = -mg\varphi + \left(-v\frac{2\delta m}{dt}\right)$$

$u = -v$

$$m\ddot{\varphi}l + mg\varphi + 2\delta m\dot{\varphi}l = 0$$ рассеивание энергии

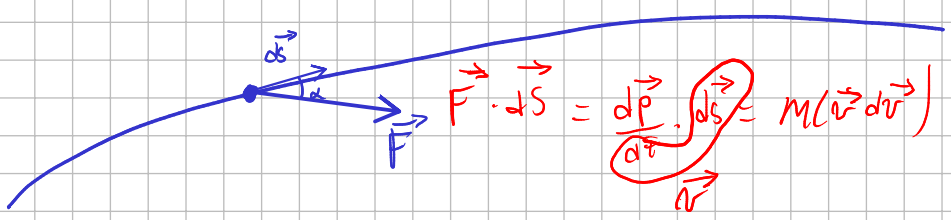
$$\ddot{\varphi} + 2\delta\dot{\varphi} + \frac{g}{l}\varphi = 0$$

решение:

коэф. затухания

$$\varphi(t) = Ae^{-\delta t}\cos\left(\sqrt{\frac{g}{l}-\delta^2}\,t + \varphi_0\right)$$

Работа и энергия



$$\vec{F}\cdot d\vec{s} = \frac{d\vec{p}}{dt}\cdot d\vec{s} = m(\vec{v}\,d\vec{v})$$

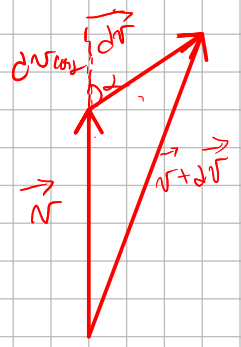
$\vec{v}$

$$\delta A = \vec{F}\cdot d\vec{s} = F\,ds\cos\alpha$$

$$A_{12} = \int_{1\to2}\delta A = \int_{1\to2}\vec{F}\cdot d\vec{s} =$$

$$= \int_{1\to2} m(\vec{v}, d\vec{v}) = \int_{1\to2} mv\,dv = \frac{mv_2^2}{2} - \frac{mv_1^2}{2}$$

$v^2 = \vec{v}^2 = (\vec{v}, \vec{v})$ $\quad 2v\,dv = 2(\vec{v}, d\vec{v})$



$\cancel{v^2} + 2v\,dv\cos\alpha + \cancel{dv^2} = \cancel{v^2} + 2v\,dv + \cancel{dv^2}$

$2v\,dv\cos\alpha = 2(\vec{v}, d\vec{v}) = 2v\,dv$

$K \equiv T = \frac{mv^2}{2}$ – кинетическая энергия

Силовое поле

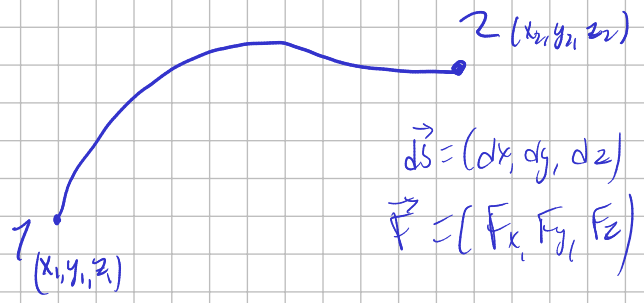
$\vec{F}(x, y, z, \cancel{t})$

стационарное поле

Опр. Поле сил потенциально $\Leftrightarrow \exists\, \Pi : F_x = -\frac{\partial \Pi}{\partial x} \quad F_y = -\frac{\partial \Pi}{\partial y} \quad F_z = -\frac{\partial \Pi}{\partial z}$

Π – U

Если $\Pi(x, y, z)$ не зависит от t, то система консервативна



$d\vec{s} = (dx, dy, dz)$

$\vec{F} = (F_x, F_y, F_z)$

Частная производная

$F(x, y, z)$

$\frac{\partial F}{\partial x} = \lim_{\Delta x \to 0} \frac{F(x+\Delta x, y, z) - F(x, y, z)}{\Delta x}$

$F = xe^{yz}$ $\quad \frac{\partial F}{\partial x} = e^{yz}, \quad \frac{\partial F}{\partial y} = xze^{yz},$

$\frac{\partial F}{\partial z} = xye^{yz}$

$dF = \frac{\partial F}{\partial x}dx + \frac{\partial F}{\partial y}dy + \frac{\partial F}{\partial z}dz$

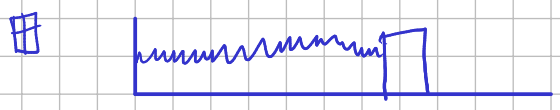
$A_{12} = \int_{1\to 2} (\vec{F}, d\vec{s}) = \int_{1\to 2} F_x dx + F_y dy + F_z dz = \int_{1\to 2} \left(-\frac{\partial \Pi}{\partial x}dx - \frac{\partial \Pi}{\partial y}dy - \frac{\partial \Pi}{\partial z}dz\right) =$

$= -\int_1^2 d\Pi = \Pi_1 - \Pi_2$ $\quad \boxed{A_{12} = \Pi_1 - \Pi_2}$

если $m = 1$ ед, то $\Pi = \Phi$ – потенциал

ЗСЭ

$\delta A = dK = -d\Pi$, т.е. $d(K + \Pi) = 0 \Rightarrow K + \Pi = const$



$E = K + \Pi$

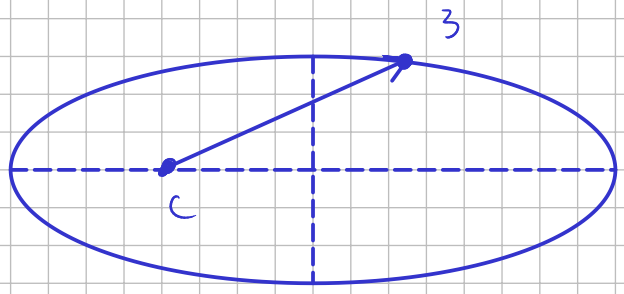
$$\vec{F}(F_x, F_y, F_z) \qquad \vec{ds}(dx, dy, dz)$$

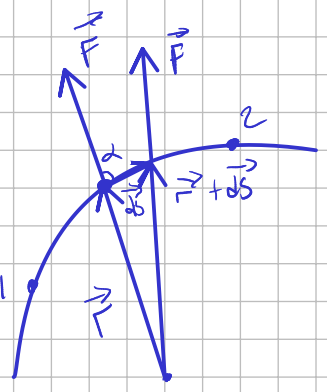
$$\delta A = (\vec{F}, \vec{ds}) = F_x dx + F_y dy + F_z dz$$

$$F_x = -\frac{\delta \Pi}{\delta x} \qquad F_y = -\frac{\delta \Pi}{\delta y} \qquad F_z = -\frac{\delta \Pi}{\delta z}$$

$$\delta A = -\frac{\delta \Pi}{\delta x} dx - \frac{\delta \Pi}{\delta y} dy - \frac{\delta \Pi}{\delta z} dz = -d\Pi \qquad \Pi(x, y, z)$$

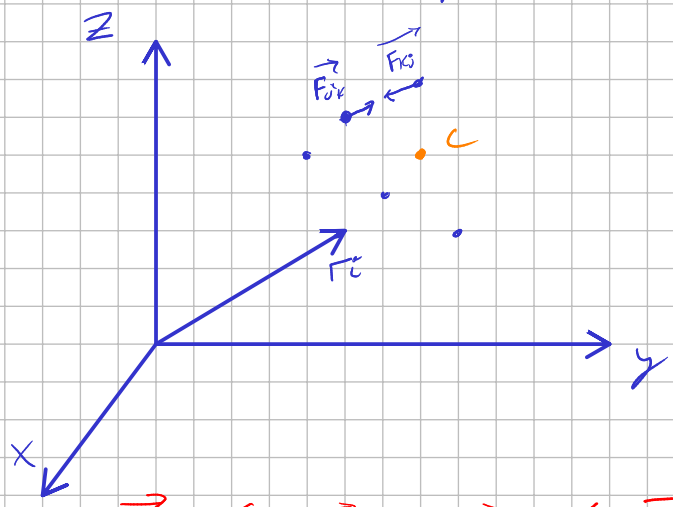
$$\delta A = d\left(\frac{mv^2}{2}\right) = dK = -d\Pi \Rightarrow d(K+\Pi) = 0 \Rightarrow K + \Pi = const$$





$$\delta A = (\vec{F}, \vec{ds}) = \underbrace{F ds \cos\alpha}_{dr} = F dr$$

$$A = \int_{1\to 2} (\vec{F}, \vec{ds}) = \int F dr = \Pi_1 - \Pi_2$$



$$\sum_{i,j} \vec{F_{ij}}^{\text{внутренних}} = 0$$

$$\sum_i \vec{p_i} = const \quad (\text{замкнутая система})$$

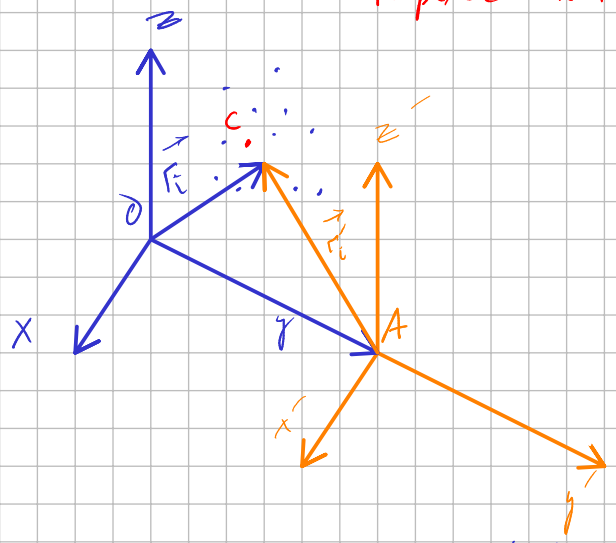
центр инерции (масс)

Опр: $\vec{r_c} = \dfrac{\sum m_i \vec{r_i}}{\sum m_i} \qquad \vec{v_c} = \dfrac{\sum m_i \vec{v_i}}{\sum m_i} \qquad \vec{a_c} = \dfrac{\sum m_i \vec{a_i}}{\sum m_i} = \dfrac{\sum \vec{F_i}}{\underbrace{\sum m_i}_{M}}$

$$M\vec{r_c} = \sum \vec{F_i} = \cancel{\sum \vec{F_i}^{\text{внутр}}}^{\,0} + \sum \vec{F_i}^{\text{внешн.}} = \sum \vec{F_i}^{\text{внешних}}$$

Теорема Кёнига

$$\text{отн. } O: \quad K = \sum_i \frac{m_i v_i^2}{2}$$



$$\vec{r}_i = \vec{r}_i' + \vec{r}_A$$

$$\vec{v}_i = \vec{v}_i' + \vec{V}_A$$

$$\vec{v}_i' = \vec{v}_i^{\text{отн}A}$$

$$K = \sum_i \frac{1}{2} m_i \left(\vec{v}_i^{\text{отн}A} + \vec{V}_A\right)^2 =$$

$$= \frac{1}{2} \sum_i m_i V_A^2 + \sum_i m_i \left(\vec{v}_i^{\text{отн}A}, \vec{V}_A\right) + \frac{1}{2} \sum_i m_i \vec{v}_i^{\text{отн}A\,2}$$

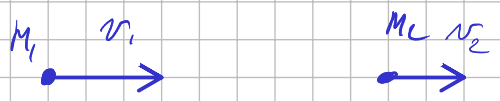
$$\sum_i m_i v_i^{\text{отн}A} = M V_C^{\text{отн}A}$$

$$K = \frac{1}{2} \sum_i m_i V_A^2 + M \vec{V}_C^{\text{отн}A} \cdot \vec{V}_A + \frac{1}{2} \sum_i m_i v_i^{\text{отн}A\,2}$$

если $\underbrace{A = C}_{\text{одна и та же точка}}$, то $\vec{V}_{C\,\text{отн}A} = 0$, $K = \frac{M V_C^2}{2} + \frac{1}{2} \sum_i m_i v_{i\,\text{отн}C}^2$

Абсолютно упругий удар.

после удара v_1', v_2'



$$\begin{cases} m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2' \\ m_1 v_1^2 + m_2 v_2^2 = m_1 v_1'^2 + m_2 v_2'^2 \end{cases}$$

$$m_1 (v_1 - v_1') = m_2 (v_2' - v_2)$$

$$m_1 (v_1^2 - v_1'^2) = m_2 (v_2'^2 - v_2^2)$$

$$v_1 + v_1' = v_2 + v_2'$$

$$v_1' = v_2 - v_1 + v_2'$$

$$m_1 v_1 + m_2 v_2 = m_1 v_2 - m_1 v_1 + m_1 v_2' + m_2 v_2'$$

$$v_2' = \frac{2 m_1 v_1 + (m_2 - m_1) v_2 - m_2 v_2 + m_2 v_2}{m_1 + m_2} = -v_2 + 2 \underbrace{\frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}}_{v_c}$$

$$v_1' = \frac{2 m_2 v_2 + (m_1 - m_2) v_1}{m_1 + m_2} = -v_1 + 2 v_c$$

Если $v_2 = 0$, $v_2' = \frac{2 m_1 v_1}{m_1 + m_2}$ $\quad v_1' = \frac{m_1 - m_2}{m_1 + m_2} v_1$

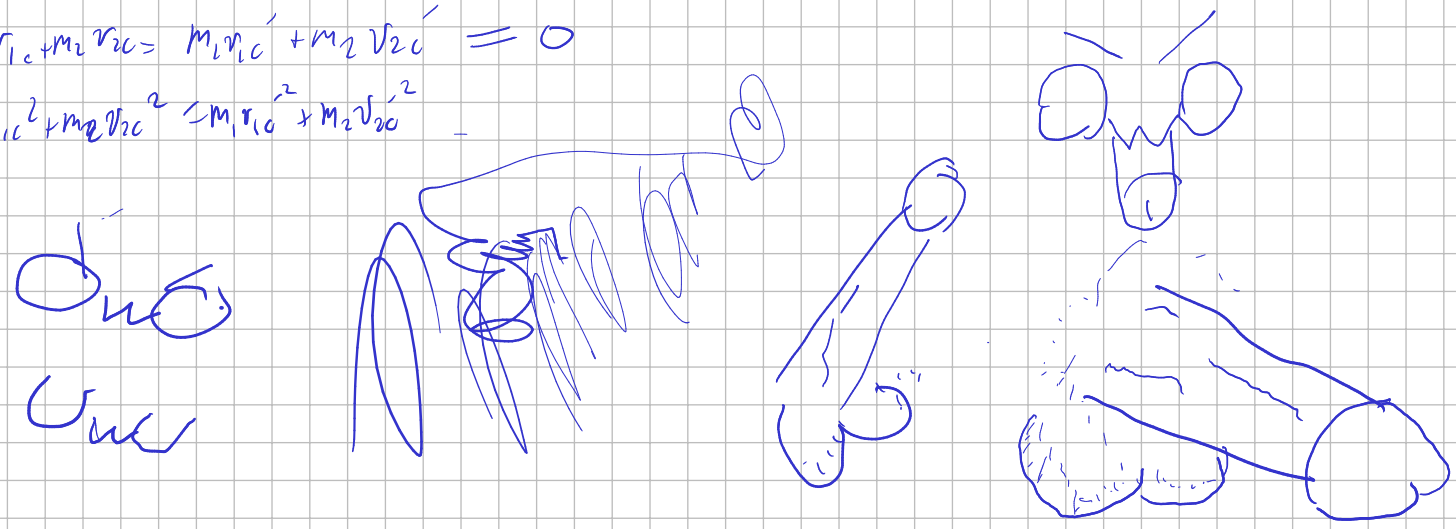
В системе центра инерции

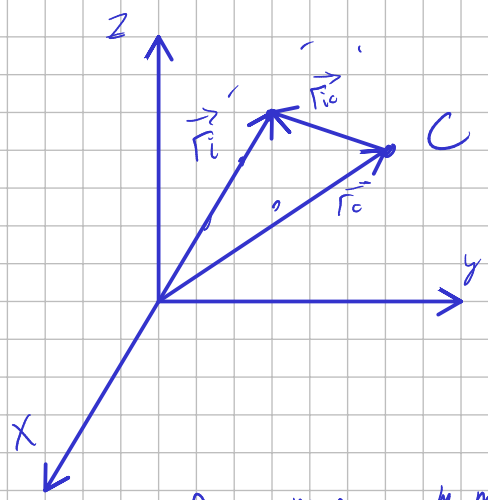
$V_c = \frac{mv_1+mv_2}{m_1+m_2}$ в ней v_{1c}, v_{2c} и v_{1c}', v_{2c}'

В СЦИ $\sum \vec{p_i} = 0$

$m_1 v_{1c} + m_2 v_{2c} = m_1 v_{1c}' + m_2 v_{2c}' = 0$

$m_1 v_{1c}^2 + m_2 v_{2c}^2 = m_1 v_{1c}'^2 + m_2 v_{2c}'^2$





$\vec{r_i} = \vec{r_c} + \vec{r_{ic}}$

$\vec{v_i} = \vec{v_c} + \vec{v_{ic}}$

$v_{1c} = v_1 - V_c = v_1 - \frac{m_1 v_1 + m_2 v_2}{m_1+m_2} = \frac{m_2}{m_1+m_2}(v_1 - v_2)$

$v_{2c} = v_2 - V_c = \frac{m_1}{m_1+m_2}(v_2 - v_1)$

$P_{1c} = m_1 v_{1c} = \frac{m_1 m_2}{m_1+m_2}(v_1 - v_2)$ $P_{2c} = m_2 v_{2c} = \frac{m_1 m_2}{m_1+m_2}(v_2 - v_1)$

μ

$P_{1c} + P_{2c} = 0$

$v_{1c}' = -v_{1c}$
$v_{2c}' = -v_{2c}$ | в СЦИ

$v_{1c}' = v_1' - V_c$
$v_{1c} = v_1 - V_c$

Упругое рассеяние тяжёлой частицы на непод. лёгкой

$v_{1c} = v_1 - V_c = v_1 \frac{m}{M+m}$

$P_{1c} = \frac{M m v_1}{M+m} = \mu v_1$

$P_{1c} + P_{2c} = 0$

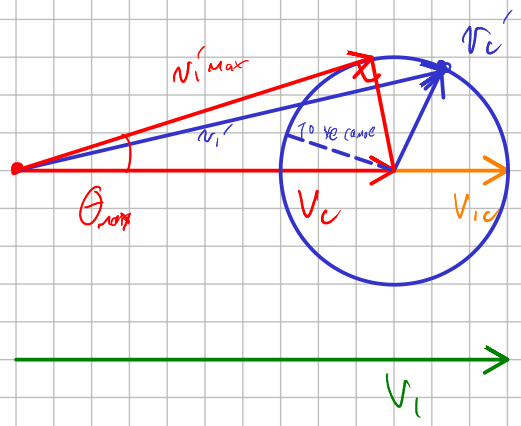
$|P_{1c}| = |P_{2c}| = P_c$

$v_{2c} = v_2 - V_c = -V_c = -\frac{M}{M+m} v_1$

$P_{2c} = -\mu v_1$

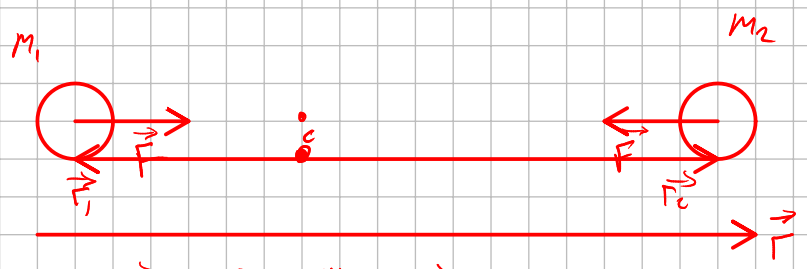
ЗСЭ: $\frac{P_c^2}{2M} + \frac{P_c^2}{2m} = \frac{P_c'^2}{2M} + \frac{P_c'^2}{2m} \iff P_c = P_c'$

Векторная диаграмма для M



$$\sin\theta_{max} = \frac{v_c'}{v_c} = \frac{K_1 \frac{m}{m+M}}{K_1 \frac{m}{m+M}} = \frac{m}{M}$$

Задача 2 тел.



$$\vec r_c = \frac{m_1 \vec r_1 + m_2 \vec r_2}{m_1+m_2} = 0$$

$$\vec r_1 = -\frac{m_2}{m_1+m_2}\vec r \qquad r_2 = \frac{m_1}{m_1+m_2}\vec r$$

$$E = \frac{m_1 \dot{\vec r}_1^{\,2}}{2} + \frac{m_2 \dot{\vec r}_2^{\,2}}{2} + \Pi(\vec r)$$

$$\Pi(\vec r) = -\frac{\gamma m_1 m_2}{r} = -\frac{\alpha}{r}$$

$$E = \frac{m_1}{2}\frac{m_2^2}{(m_1+m_2)^2}\dot{\vec r}^{\,2} + \frac{m_2}{2}\frac{m_1^2}{(m_1+m_2)^2}\dot{\vec r}^{\,2} - \frac{\gamma m_1 m_2}{r} =$$

$$= \frac{m_1 m_2}{2(m_1+m_2)}\left(\frac{m_2}{m_1+m_2} + \frac{m_1}{m_1+m_2}\right)\dot{\vec r}^{\,2} - \frac{\gamma m_1 m_2}{r} =$$

$$E = \frac{\mu \dot{\vec r}^{\,2}}{2} - \frac{\gamma m_1 m_2}{r} = \underline{\underline{const}} \qquad \mu = \frac{m_1 m_2}{m_1+m_2} = const$$

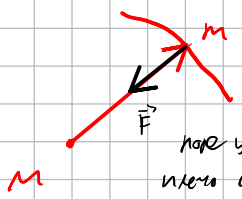
интеграл движения

рассматриваем $m_1 - M$, $m_2 - m$, $M \gg m$, тогда $\mu \simeq m$

$$E = \frac{m v^2}{2} + \Pi(r)$$

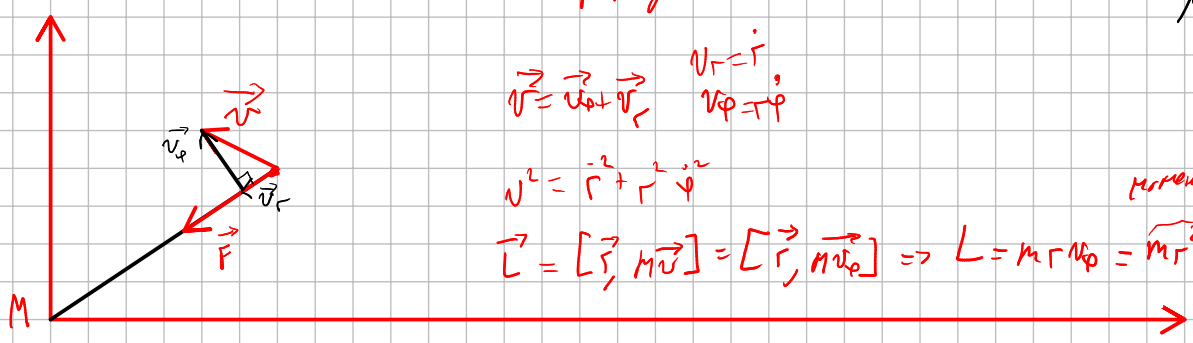
$$\frac{d\vec L}{dt} = \vec M = 0 \implies \vec L = \underline{\underline{const}}$$

интеграл движения



поле центральных сил
плечо сил $= 0$

$$\vec M = [\vec r, \vec F] = 0$$



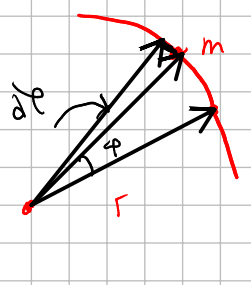
$$\vec v = \vec v_\varphi + \vec v_r \qquad v_r = \dot r, \quad v_\varphi = r\dot\varphi$$

$$v^2 = \dot r^2 + r^2\dot\varphi^2$$

момент инерции

$$\vec L = [\vec r, m\vec v] = [\vec r, m\vec v_\varphi] \implies L = m r v_\varphi = \underbrace{m r^2}_{I}\underbrace{\dot\varphi}_{\omega} \qquad L = I\omega = const$$

$\dot\varphi = \frac{L}{mr^2}$ $\qquad E = \frac{m\dot r^2}{2} + \frac{m r^2 \dot\varphi^2}{2} + \Pi(r)$



$dS = \frac{1}{2} r \cdot r d\varphi \,|: dt \qquad \frac{dS}{dt} = \frac{1}{2} r^2 \frac{d\varphi}{dt} = \frac{r^2\dot\varphi}{2} = \frac{L}{2m} = const$

Это 2-й закон Кеплера.

$E = \frac{m\dot r^2}{2} + \frac{\cancel{mr^2}}{2} \cdot \frac{L^2}{m^2 r^{\cancel{4}}} + \Pi(r) \qquad E = \frac{m\dot r^2}{2} + V(r)$

$$E = \frac{m\dot r^2}{2} + \underbrace{\frac{L^2}{2mr^2} + \Pi(r)}_{V(r)}$$

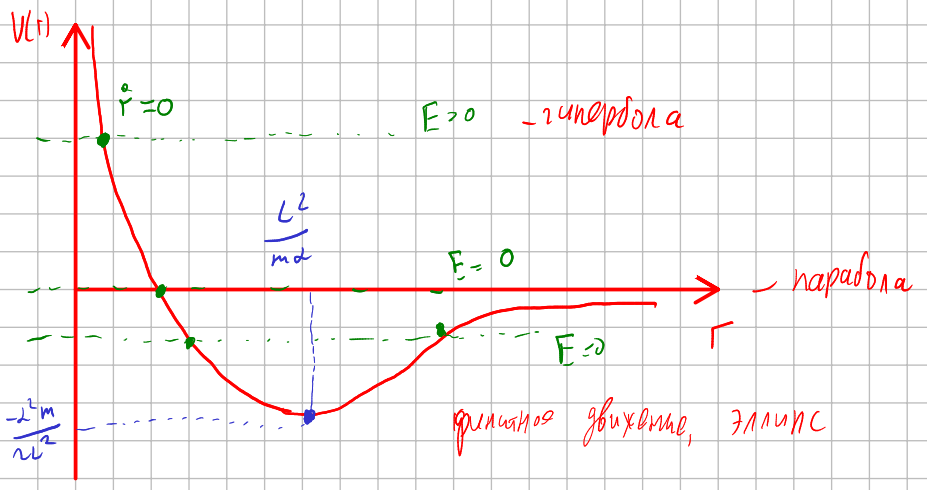
$V(r)$ - эфф. пот. энергия

$$\dot r = \sqrt{\frac{2}{m}\left(E - \Pi(r)\right) - \left(\frac{L}{mr}\right)^2} = A(r)$$

$\frac{dr}{A(r)} = dt \qquad \frac{d\varphi}{dt} = \frac{L}{mr^2} \qquad d\varphi = \frac{L}{mr^2} dt \qquad d\varphi = \frac{L}{mr^2}\frac{dr}{A(r)}$

$\varphi = \int \frac{L\, dr}{mr^2 A(r)} \qquad t = \int \frac{dr}{A(r)}$

$E = \frac{m\dot r^2}{2} + V(r) \qquad V(r) = \frac{L^2}{2mr^2} - \frac{\gamma M m}{r}$, $\gamma M m = \alpha$



в т., где $\dot r = 0$, $\frac{L^2}{2mr^2} - \frac{\alpha}{r} = V(r) = E$

$L^2 - \gamma M m\, 2mr = E\, 2mr^2$

$$r^2 + \frac{\gamma M m}{E} r - \frac{L^2}{2mE} = 0$$

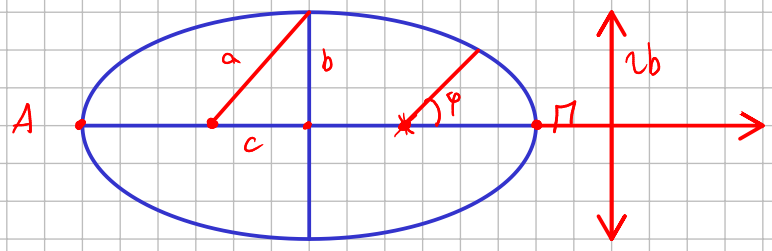
для случая $\Pi(r) = -\frac{\alpha}{r}$

$$\varphi = \arccos \frac{\frac{L^2}{m\alpha r} - 1}{\sqrt{1 + \frac{2L^2}{m\alpha^2} E}} + C \qquad p = \frac{L^2}{m\alpha}, \quad e = \sqrt{\dots} \quad \text{— эксцентриситет}$$

$$\varphi = \arccos \frac{\frac{p}{r} - 1}{e} + C$$

при таком отсчёте угла $C = 0$

$$r = \frac{p}{1 + e\cos\varphi}$$



$e = \frac{c}{a}$

$e < 1$ — эллипс

$e = 1$ — парабола

$e > 1$ — гипербола

$$r^2 + \frac{\gamma M m}{E} r - \frac{L^2}{2mE} = 0$$

$$r_1 + r_2 = -\frac{\gamma M m}{E} = 2a \qquad r_1 r_2 = -\frac{L^2}{2mE} = b^2$$

$$b = \frac{L}{\sqrt{-2mE}}$$

$$e = \frac{c}{a} = \frac{\sqrt{a^2 - b^2}}{a} = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 + \frac{2L^2E}{\gamma^2 M^2 m^2}}$$

Круговая орбита.

$$a = b = R = -\frac{\gamma M m}{E} \qquad \Pi = -\frac{\gamma M m}{R} = 2E, \qquad E = K + \Pi$$

$$K = -\frac{\Pi}{2} \neq -E$$

$$\frac{m v_{кр}^2}{R} = \frac{\gamma M m}{R^2} \qquad v_{кр} = \sqrt{\frac{\gamma M}{R}} = \sqrt{\frac{g_0 R_3^2}{R}} \qquad v_I = \sqrt{g_0 R_3} \approx 7{,}9 \ \frac{км}{с}$$

$(R = R_3)$

Парабола $E = 0$ $\qquad \frac{m v_{пар}^2}{2} = \frac{\gamma M m}{R} \qquad v_{пар} = \sqrt{\frac{2\gamma M}{R}} = \sqrt{2}\, v_{кр}$

$$v_{II} = \sqrt{2}\, v_I \approx 11{,}2 \ \frac{км}{с}$$

Для Солнца без учёта Земли

$v_{II c} = \sqrt{2} \cdot 29{,}8 \approx 42{,}1 \ \frac{км}{с}$

Период обращения

$\frac{dS}{dt} = \frac{L}{2m}$ $\qquad 2m\,dS = L\,dt$

$2m S_{эл} = LT$

$2m\pi ab = LT$

$T = \frac{2m\pi ab}{L} = \pi\gamma M_m \sqrt{\frac{m}{2|E|^3}}$

$r = \frac{p}{1+\cos\varphi}$ $\qquad \frac{dS}{dt} = \frac{1}{2} r^2 \dot{\varphi} = \frac{L}{2m}$

$2m\,dS = L\,dt \Rightarrow 2m S_{эл} = LT$

$T = \frac{2m\pi ab}{L} = \frac{2\pi m}{L} \cdot \frac{\gamma M_m}{2|E|} \cdot \frac{L}{\sqrt{2m|E|}} = \pi\gamma M_m \sqrt{\frac{m}{2|E|^3}}$

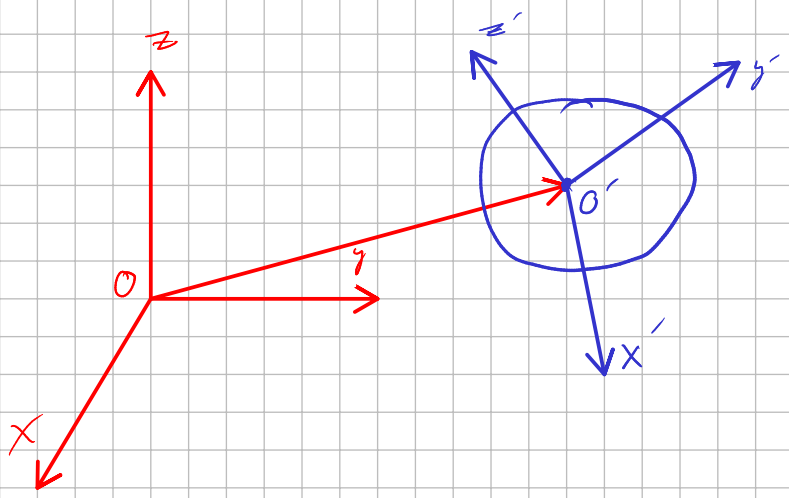
$a = -\frac{\gamma M_m}{2E}$

$b = \frac{L}{\sqrt{-2mE}}$

$\frac{T^2}{a^3} = \frac{\pi^2\gamma^2 M^2 m^2 m 8 E^3}{2E^3\gamma^3 M^3 m^3} = \frac{4\pi^2}{\gamma M} = const$

$\frac{T^2}{a^3} = \frac{4\pi^2}{\gamma(M+m)}$ – на самом деле

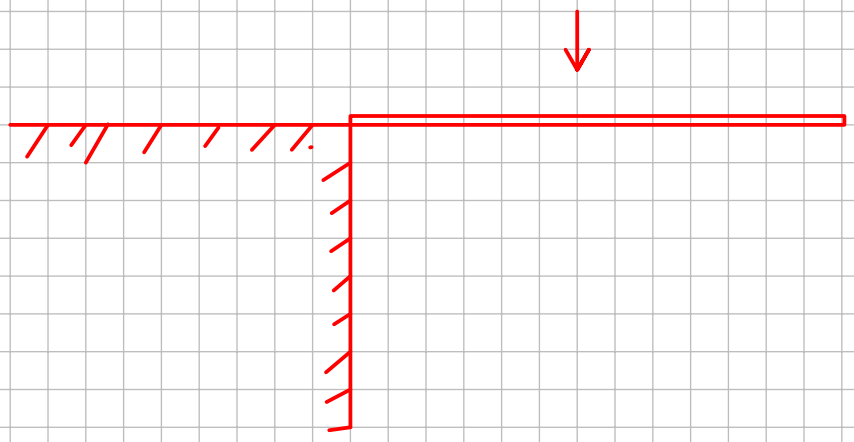
Движение твёрдого тела

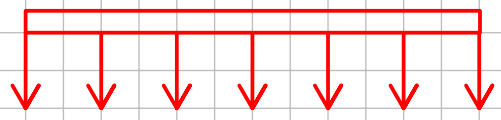


$$m\frac{d\vec{V}}{dt} = \sum \vec{F_i}$$

$$\frac{d\vec{L_0}}{dt} = \sum \vec{M_i}$$

У твёрдого тела 6 степеней свободы - положение в пространстве описывается 6-ю числами.





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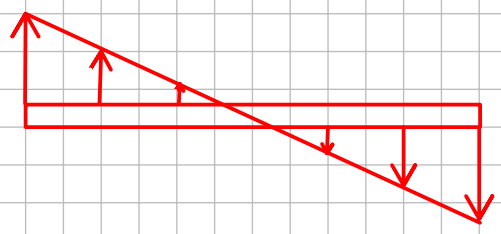

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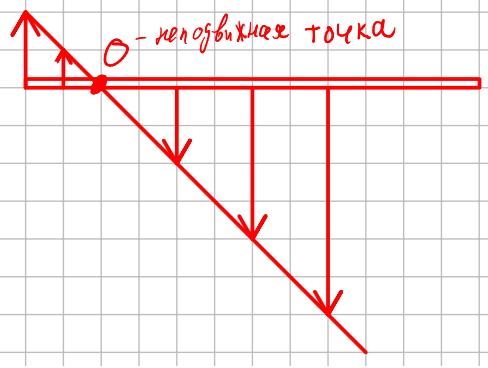
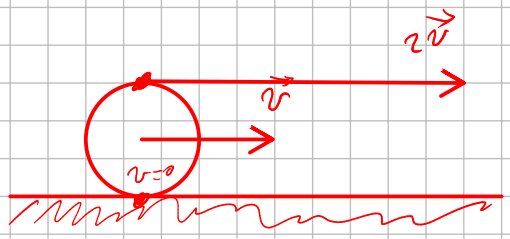
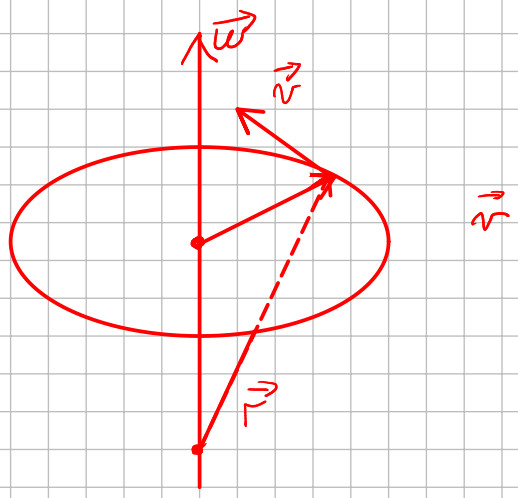


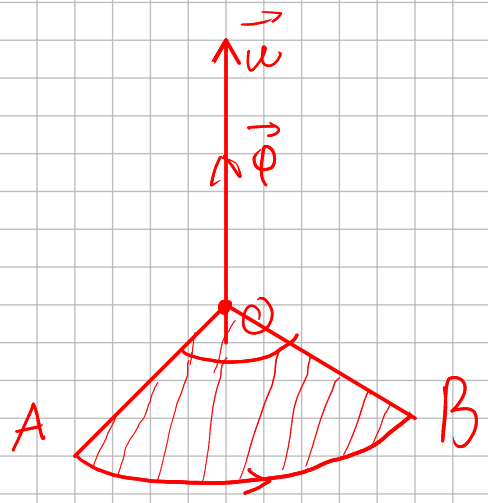


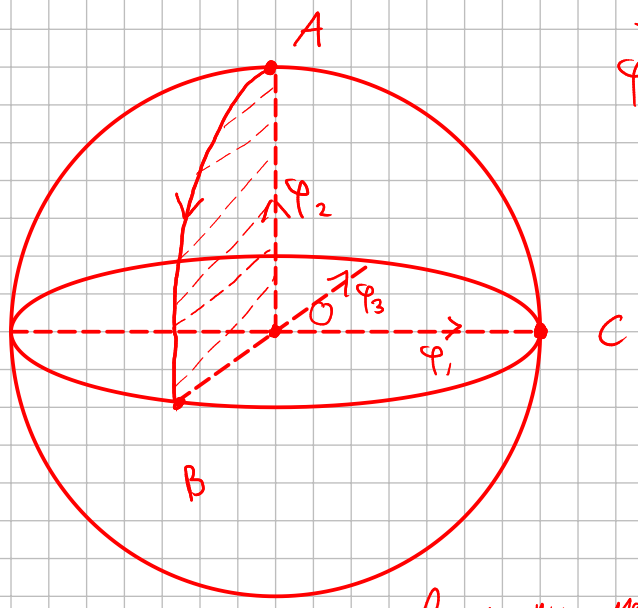


$\vec{v} = [\vec{\omega}, \vec{r}]$ $\quad \vec{\varphi} = \int \vec{\omega}\, dt$

$\vec{\varepsilon} = \dot{\vec{\omega}}$

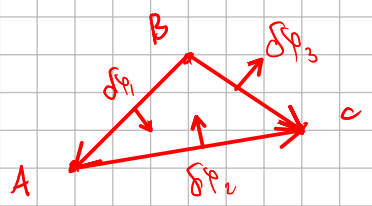
$\vec{v}$





$\vec{\varphi}_3 = \vec{\varphi}_1 + \vec{\varphi}_2$

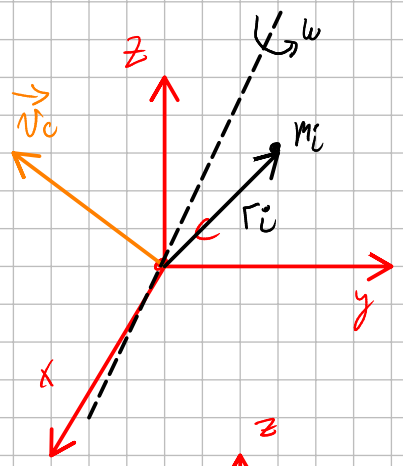
но верно при малых углах



$\delta\varphi_3 = \delta\varphi_1 + \delta\varphi_2$

Кёниг

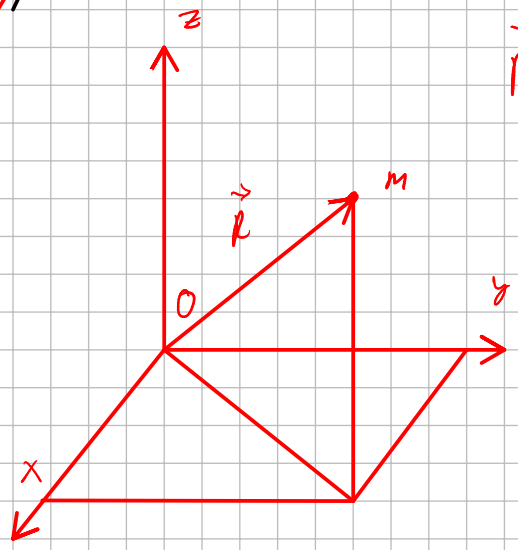
$$K = \frac{M v_c^2}{2} + \sum_i \frac{m_i v_{отн_i}^2}{2}$$



$$\vec{v}_{i\,отн} = [\vec{r}_i ; \vec{\omega}]$$

$$v_{i\,отн} = \omega \rho_i$$

$$\sum_i \frac{m_i v_{i\,отн}^2}{2} = \underbrace{\sum_i \frac{m \rho_i^2}{2}}_{\text{момент инерции}} \omega^2 = \frac{I \omega^2}{2}$$



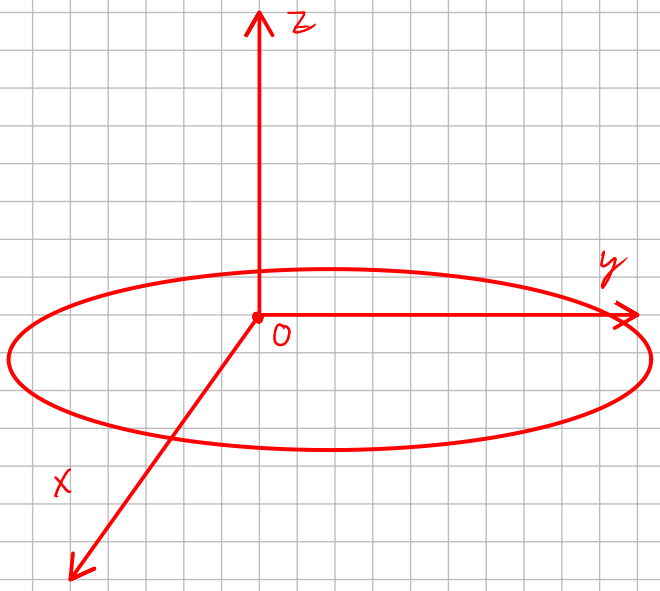
$\vec{R}(x, y, z)$

$I_x = m(y^2+z^2)$

$I_y = m(x^2+z^2)$

$I_z = m(x^2+y^2)$

$I_x + I_y + I_z = 2m(x^2+y^2+z^2) = 2mR^2$



$I_x + I_y + I_z = 2I_z$

$I_x + I_y = I_z$

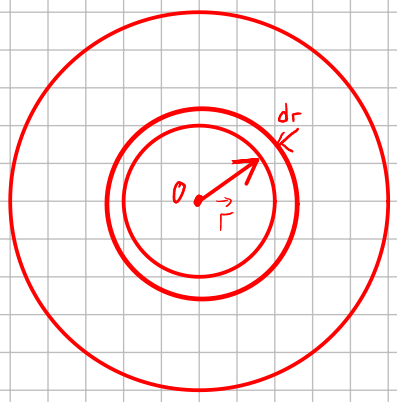
$I_x = I_y$, если диск

$2I_x = I_z$

$I_x = \frac{I_z}{2}$, $I_z = MR^2$

$\Rightarrow I_x = I_y = \frac{MR^2}{2}$

Диск (цилиндр)



$$dI_z = dm\, r^2$$

$$\frac{dm}{m} = \frac{2\pi\, r\, dr\, \cancel{\ell\rho}}{\pi R^2 \cancel{\ell\rho}} = \frac{2r\,dr}{R^2}$$

$$I_z = \int dI_z = \frac{2m}{R^2}\int_0^R r^3\,dr = \frac{mR^2}{2}$$

$$I_x = I_y = \frac{mR^2}{4}$$

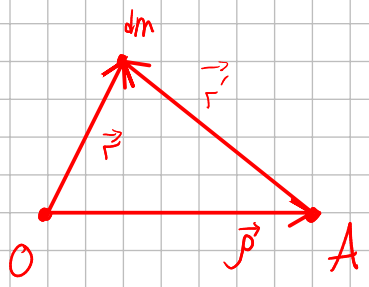
Сфера: $I_0 = \frac{2}{3} mR^2$

Шар: $I_ш = \frac{2}{5} mR^2$

Стержень относительно конца: $I = \frac{ml^2}{3}$

отн. ц. масс $I = \frac{ml^2}{12}$

Теорема Гюйгенса штейнера



$$\vec{r}' = \vec{r} - \vec{\rho}$$

$$r'^2 = r^2 + \rho^2 - 2(r, \rho)$$

$$\vec{r}_c = \int \frac{\vec{r}\,dm}{M}$$

$$I_A = \int dm\, r'^2 = \underbrace{\int r^2 dm}_{I_0} + \underbrace{\int \rho^2 dm}_{\rho^2 M} - 2\Big(\vec{\rho}\underbrace{\int \vec{r}\,dm}_{M\vec{r}_c}\Big) =$$

$$= I_0 + M\rho^2 - 2M(\vec{\rho}, \vec{R}_c),$$ если O – ц.м., то $\vec{R}_c = 0$,

тогда $I_A = I_c + M\rho^2$